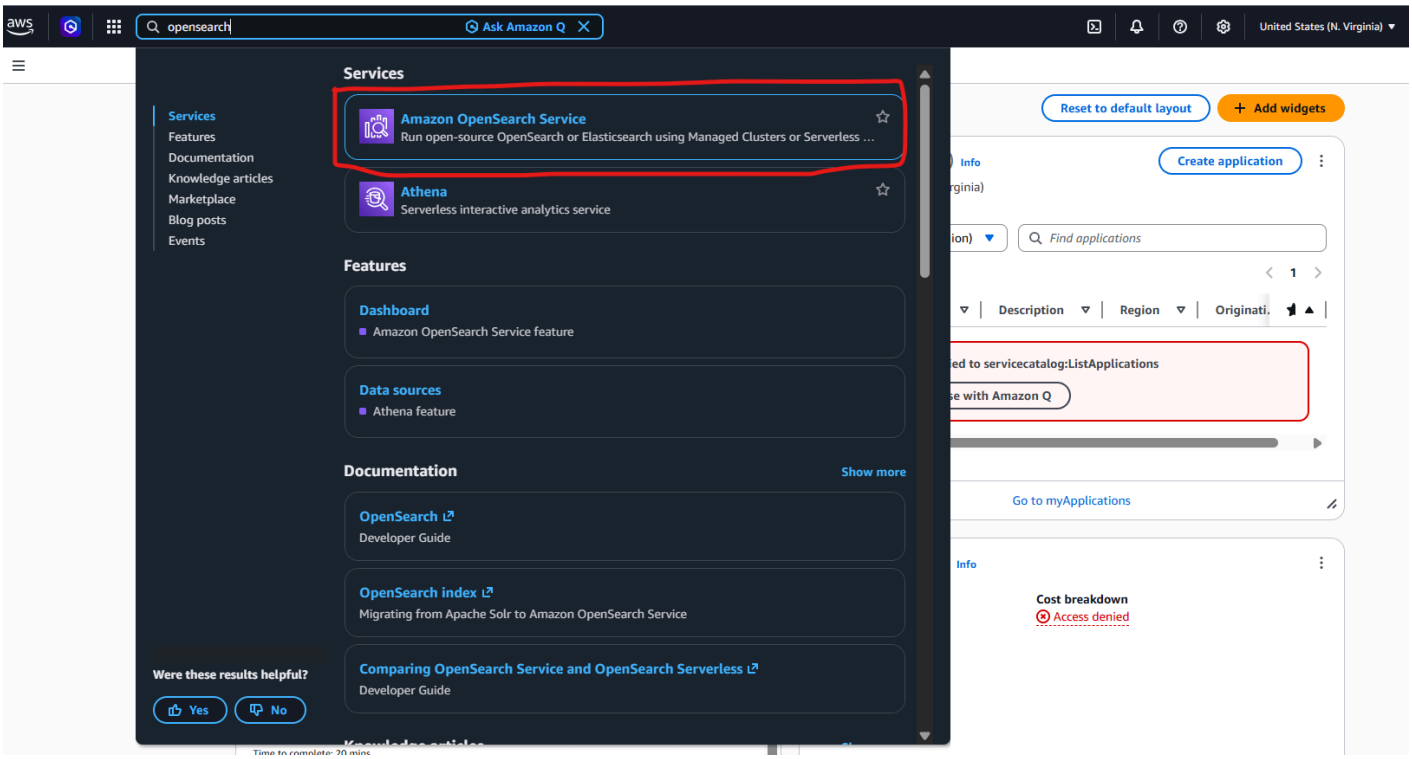
AWS-OpenSearch-ServerLess-Lambda

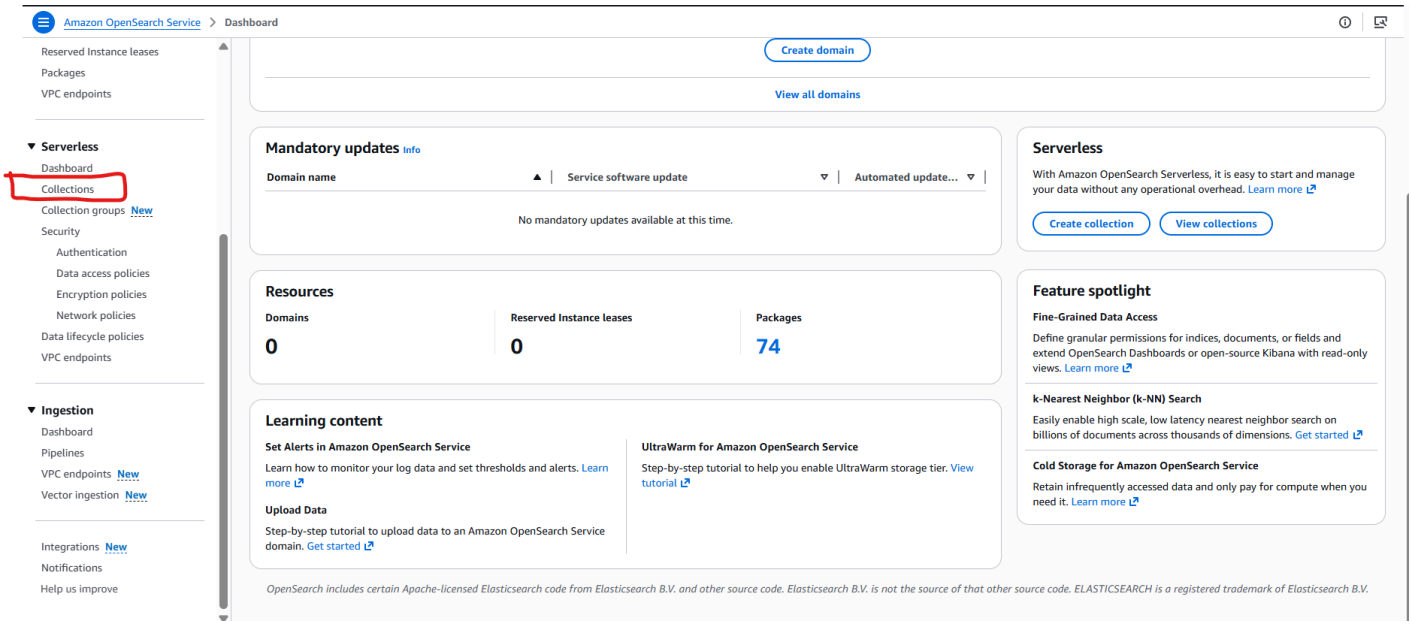
In This lab we will create a Lambda function that will sends the data to AWS open search server less collection.

Create a Open Search Server-less collection

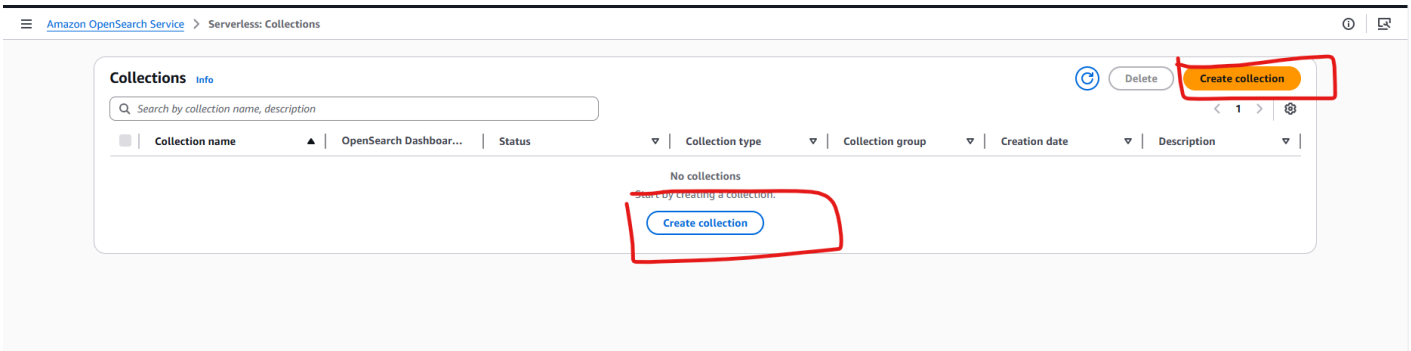
Go to AWS Console



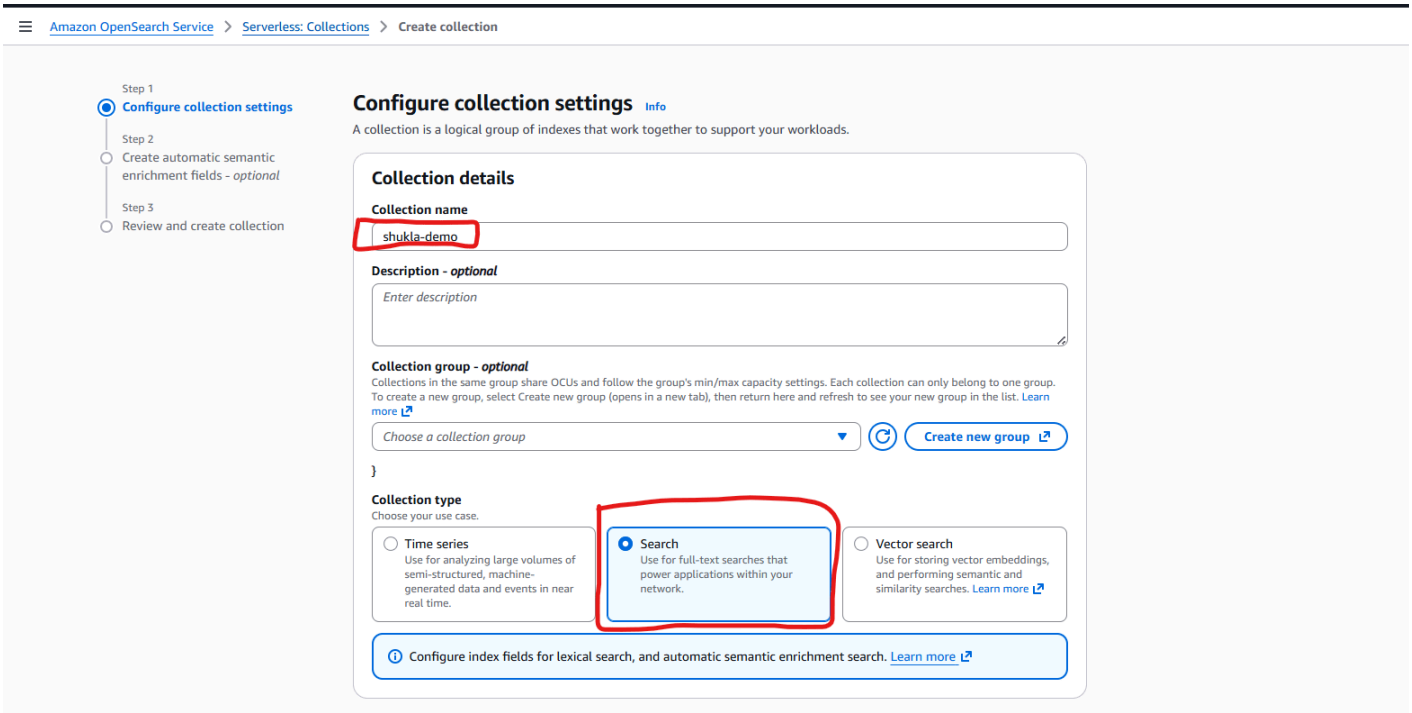
From the left panel, Select "collections".



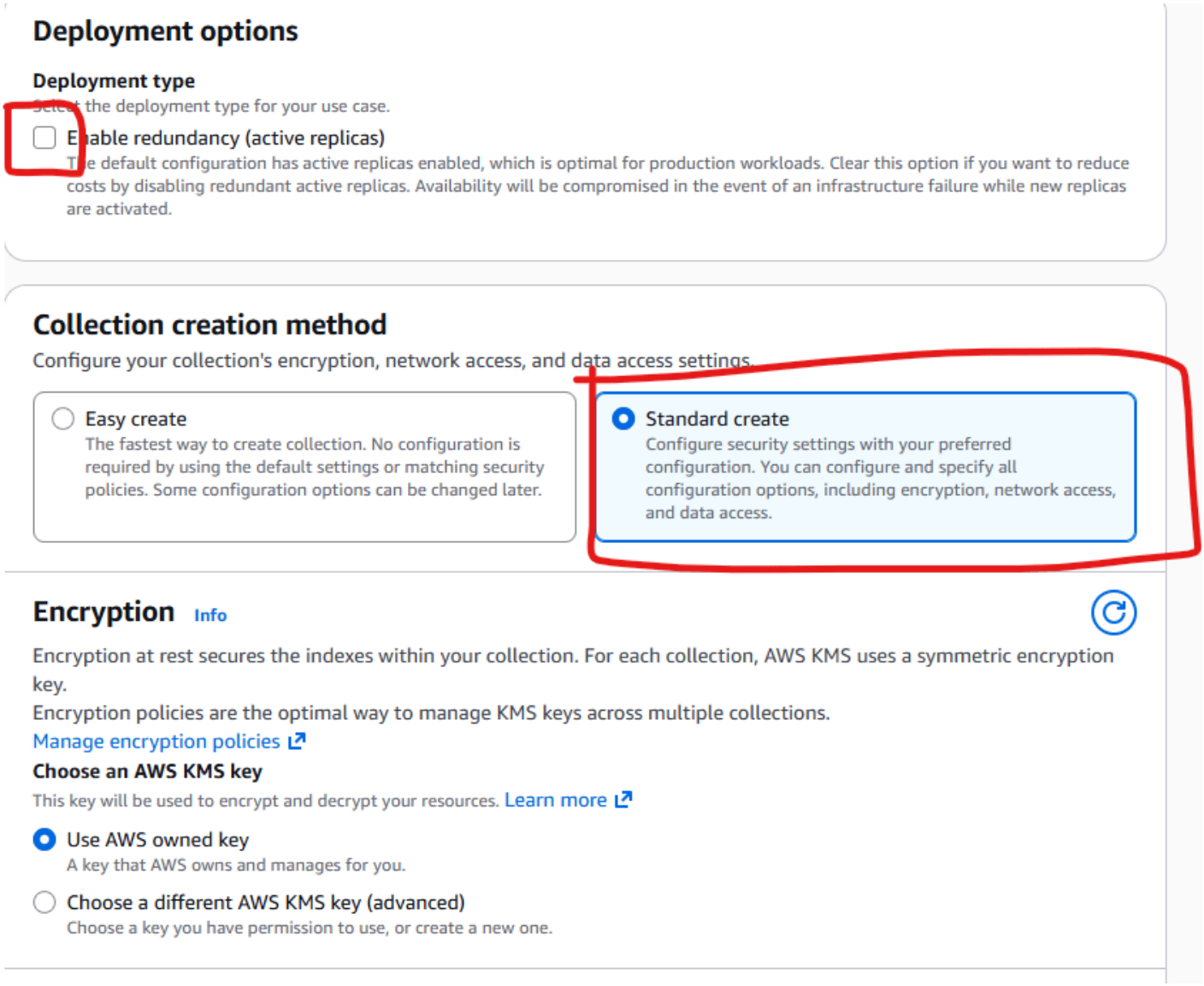
Click on Create Collection.



Provide a name to Collection and Select "search"



un select the redundancy check box and select standard create.



Select Network access public. Check both resource type and click next.

Network access settings Info

Choose internet or VPC access. To enable VPC access, we use private IP addresses from your VPC, which provides an inherent layer of security. Collection VPC endpoints are created and managed solely within OpenSearch Serverless and are separate from domain VPC endpoints. The optimal way to manage network access and VPC endpoints across multiple collections is using network access policies.
View and manage network access policies

Access type

You can configure the same or different network settings for your collection endpoint and its corresponding OpenSearch Dashboards endpoint.

Access collections from

- (●) Public
- ( ) Private (recommended)

Resource type

Configure network access separately for a collection's OpenSearch endpoint and its corresponding OpenSearch Dashboards endpoint. If you specify access for a collection's Dashboards endpoint, the settings must be the same as its collection endpoint, otherwise Dashboards won't function properly.

- [x] Enable access to OpenSearch endpoint
- [x] Enable access to OpenSearch Dashboards

Tags - *optional*

You can add tags to describe your collection. A tag consists of a case-sensitive key-value pair. For example, you can define a tag with a key-value pair of Environment Name-Development.

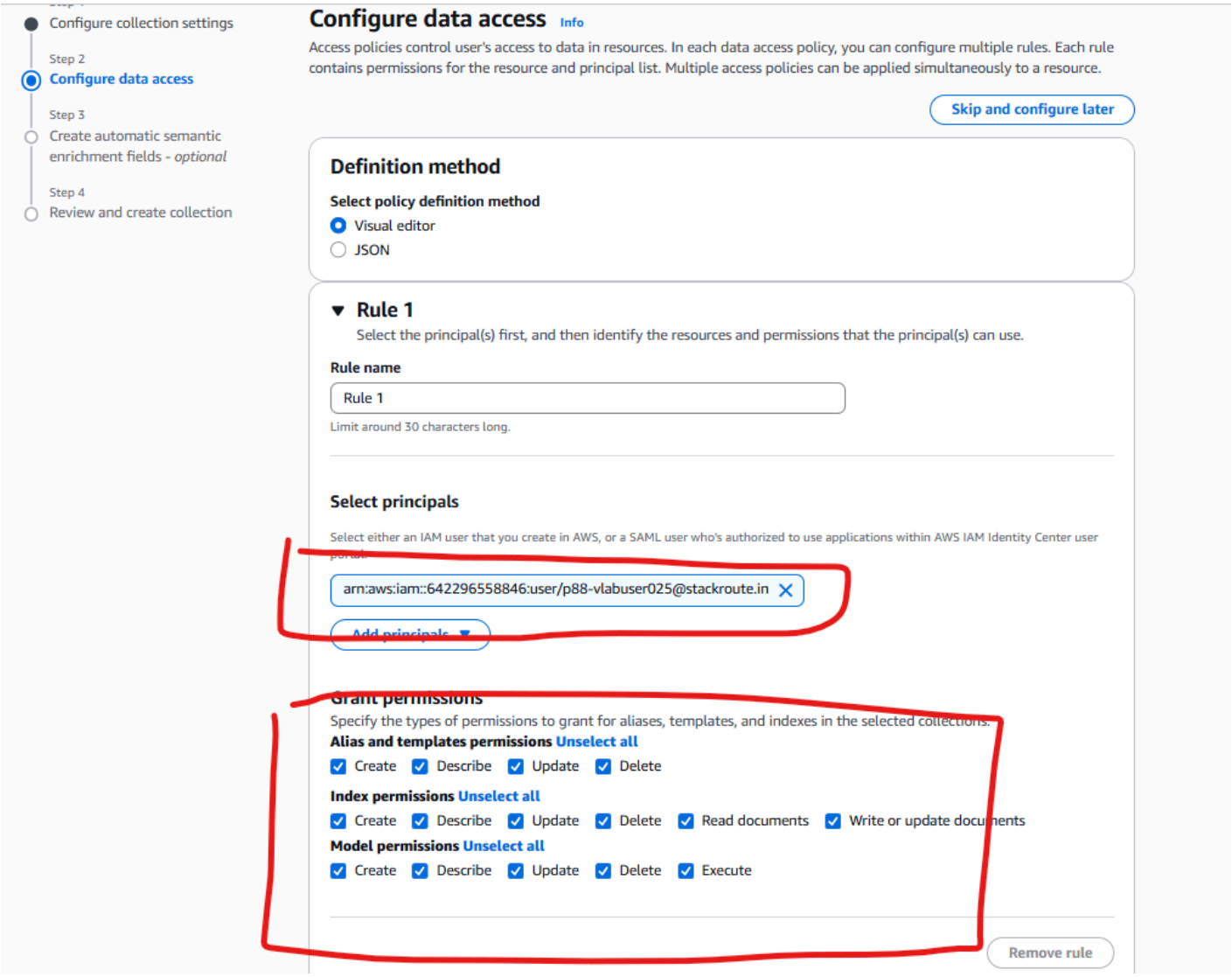
No tags associated with this collection

Add new tag

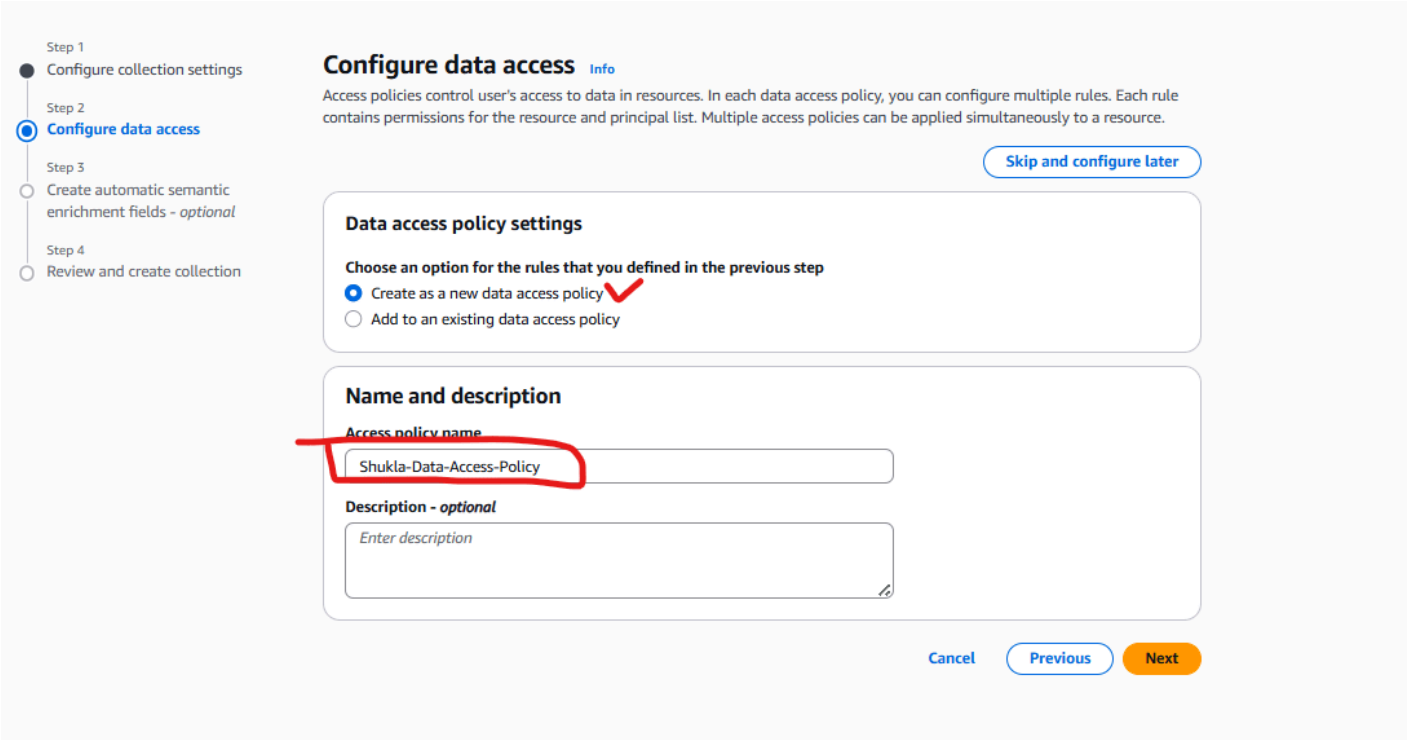
You can add 50 more tags.

Cancel Next

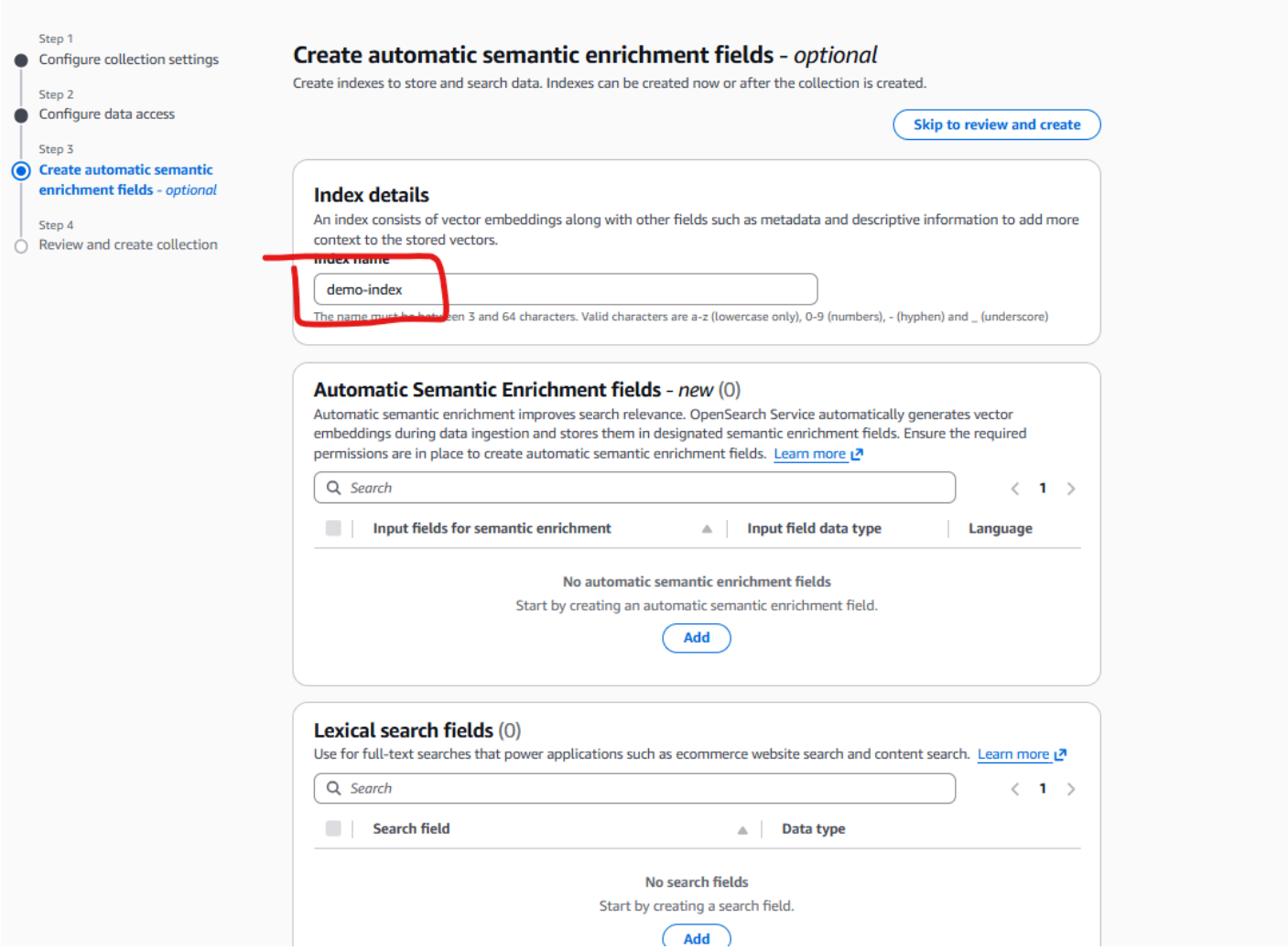
Select appropriate data access permissions.



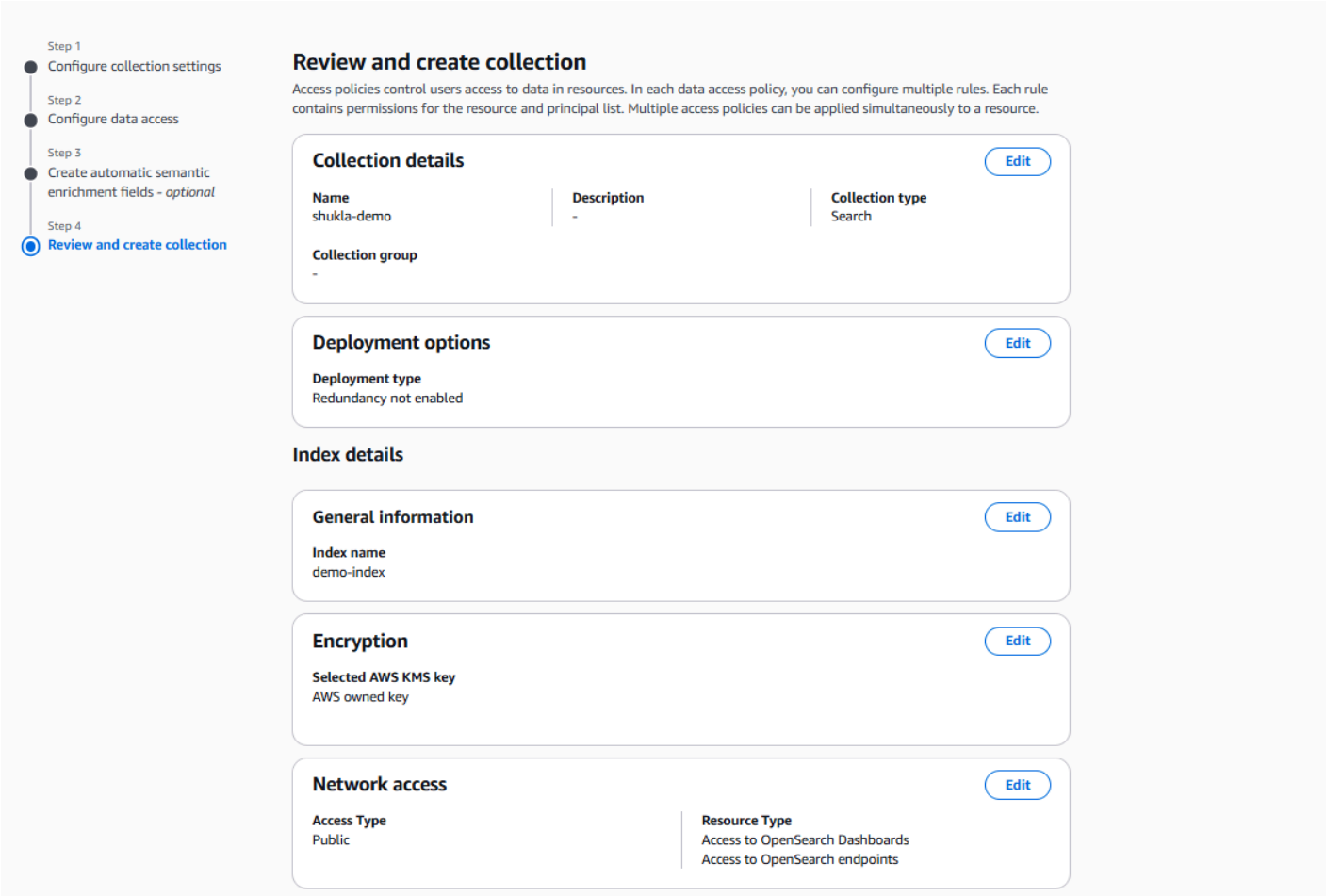
Save it as a new policy and click next.



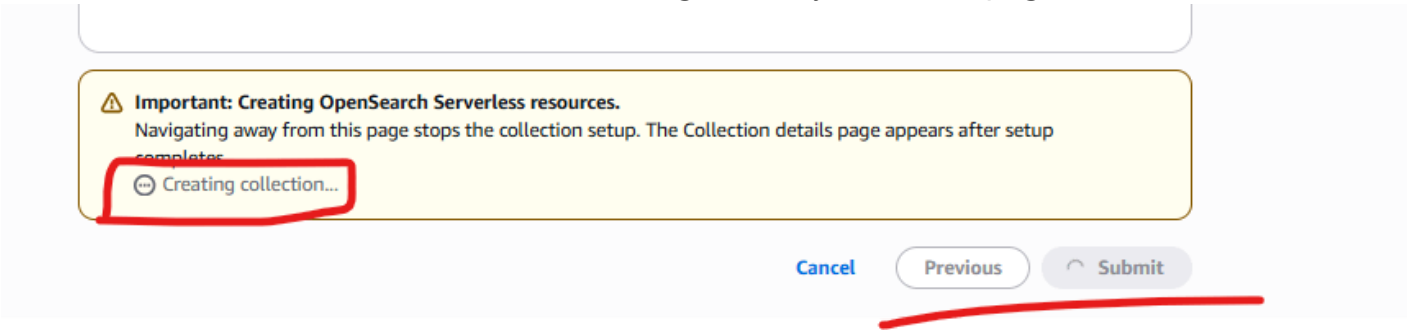
Name the Index as "" demo-index



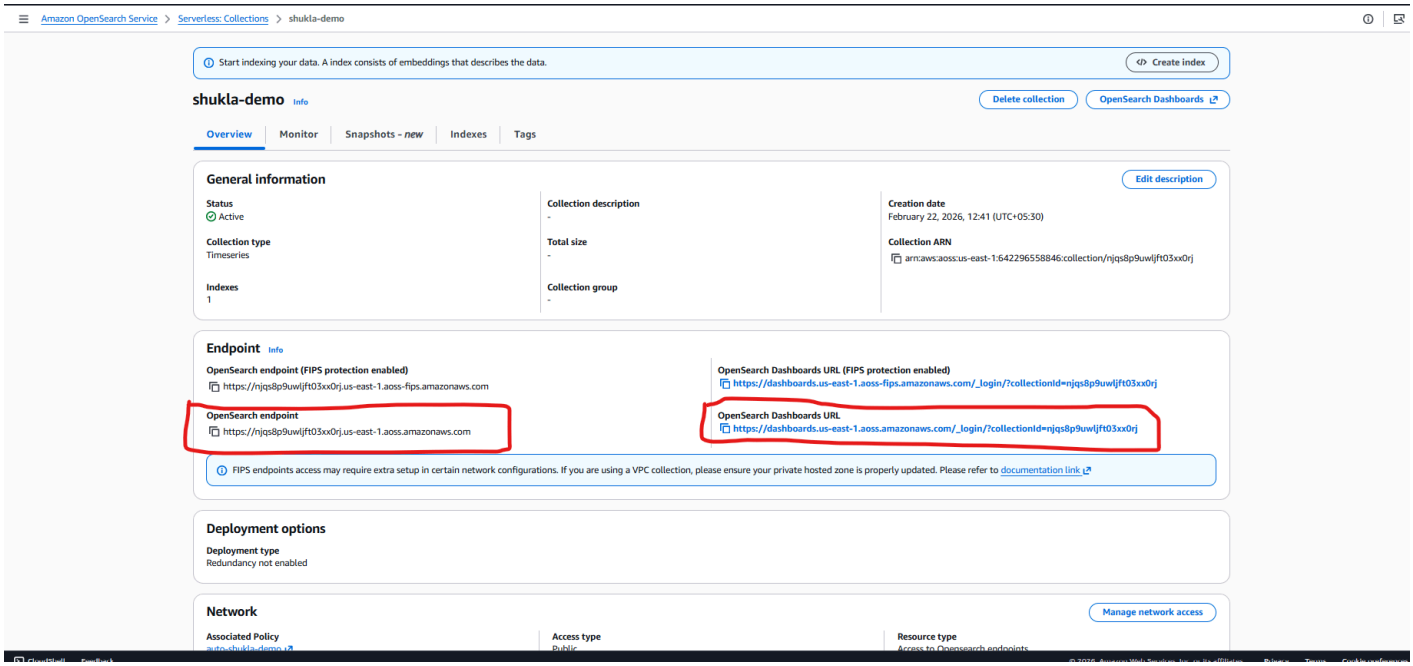
Review and click Submit.



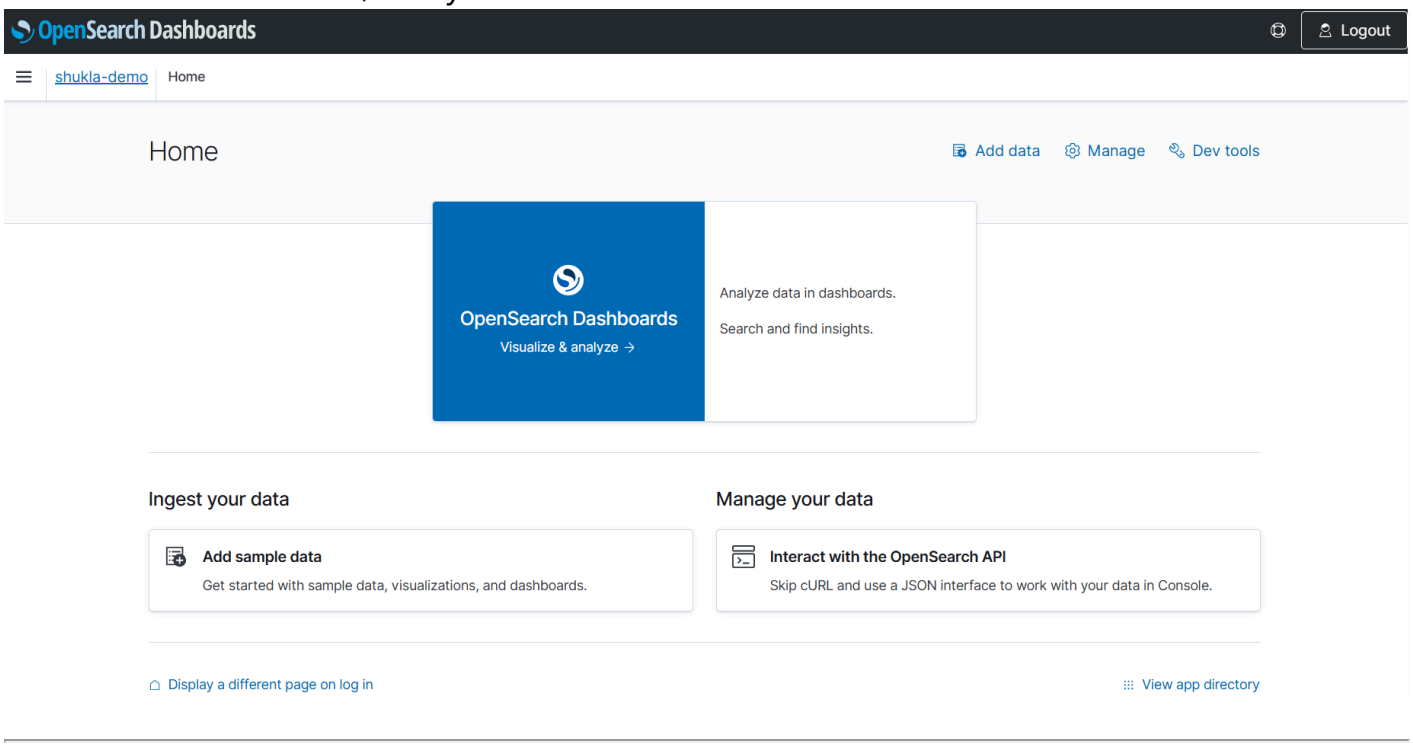
wait for collection to be created. Do not navigate away from this page.



Once your open search collection will be ready screen will look like . Note down the open search endpoint. and dashboard url. we will need this later on.



Click on Dashboard Url , and you will see the dashboard.

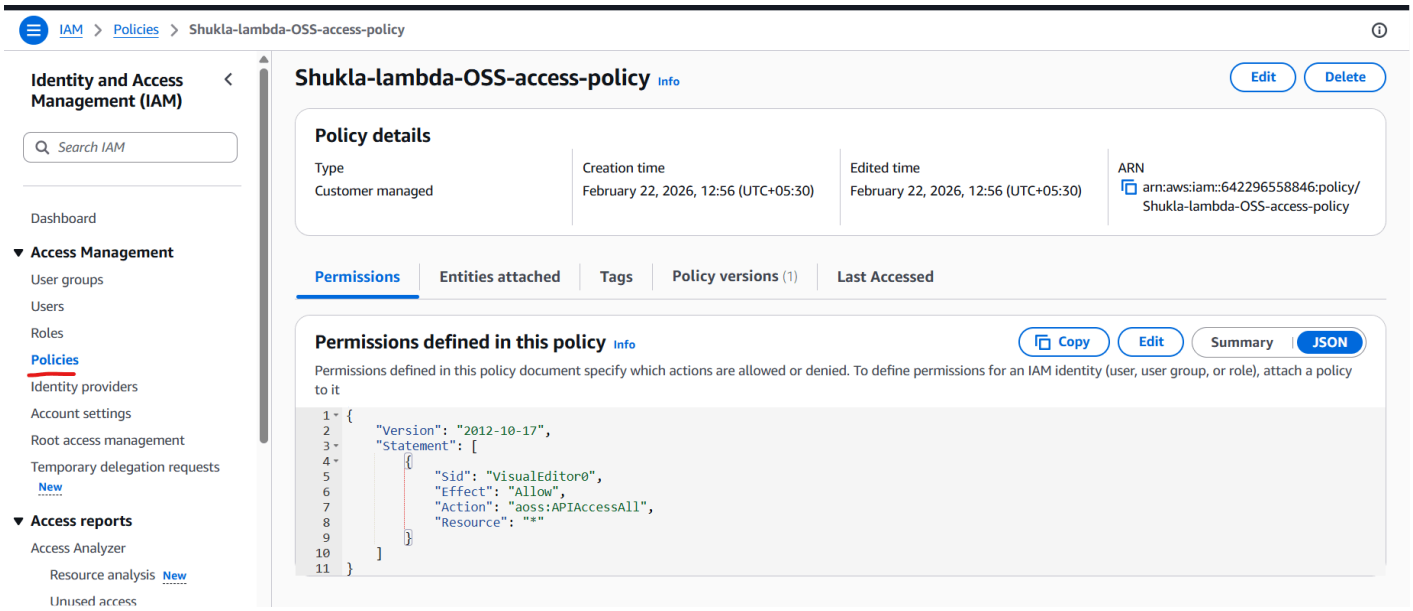


Create Lambda Function that will send Data o this open search

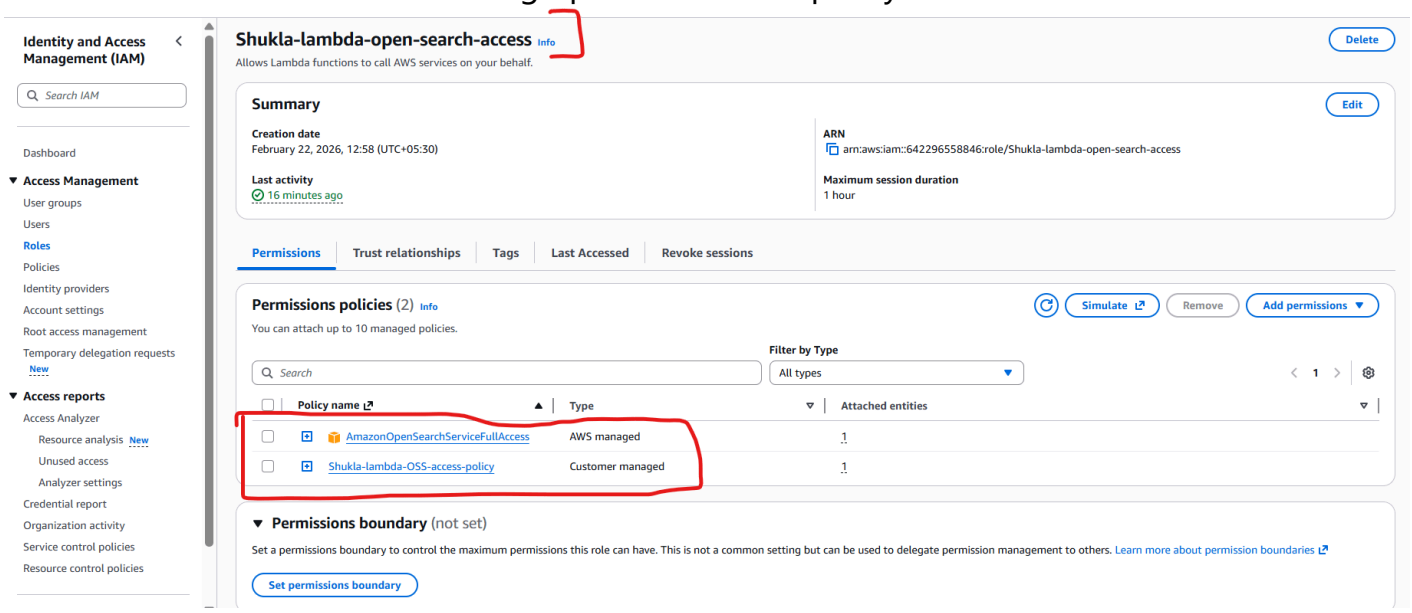
First Create a IAM Role that Lambda will use to connect with AWS opensearch collection.

Create a Policy Name "Shukla-lambda-OSS-access-policy".

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": "aoss:APIAccessAll",
            "Resource": "*"
        }
    ]
}
```



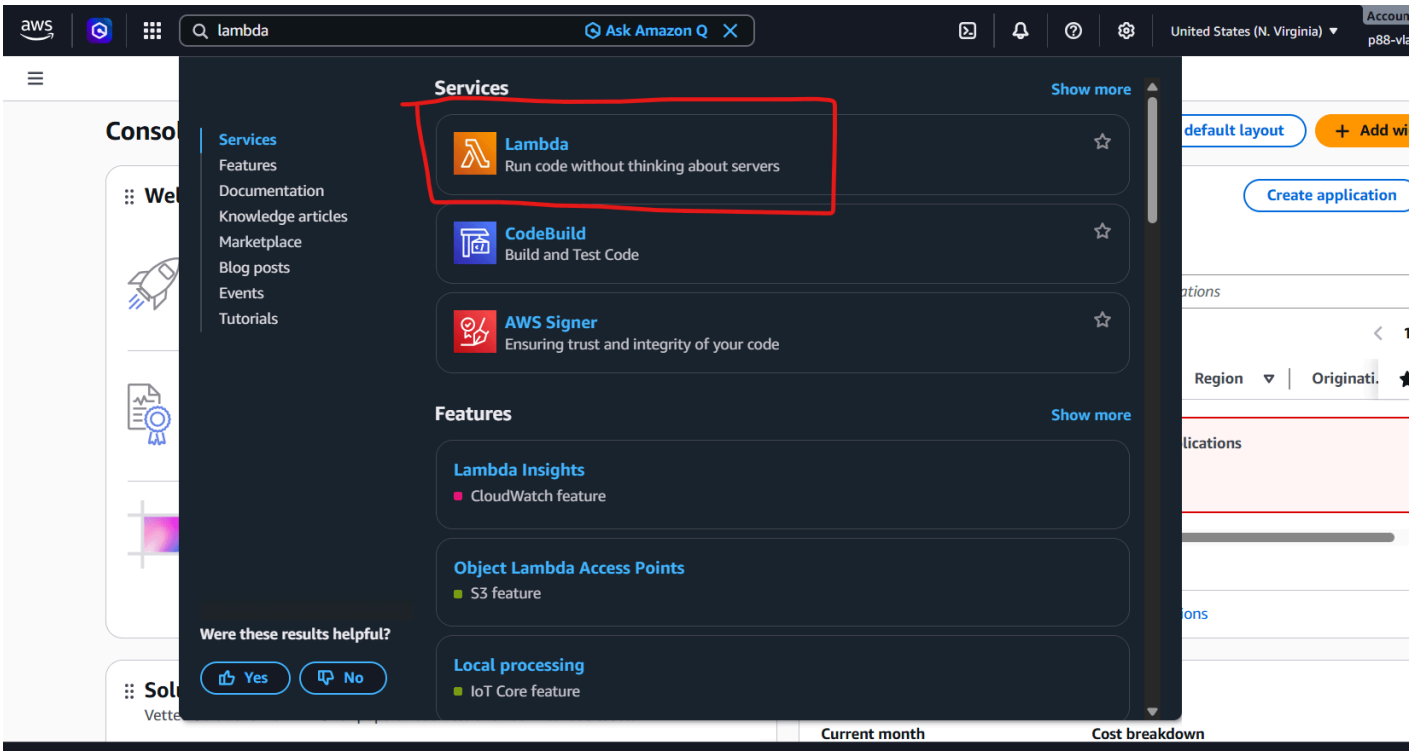
Create a Role for Lambda and assign permission and policy on it.

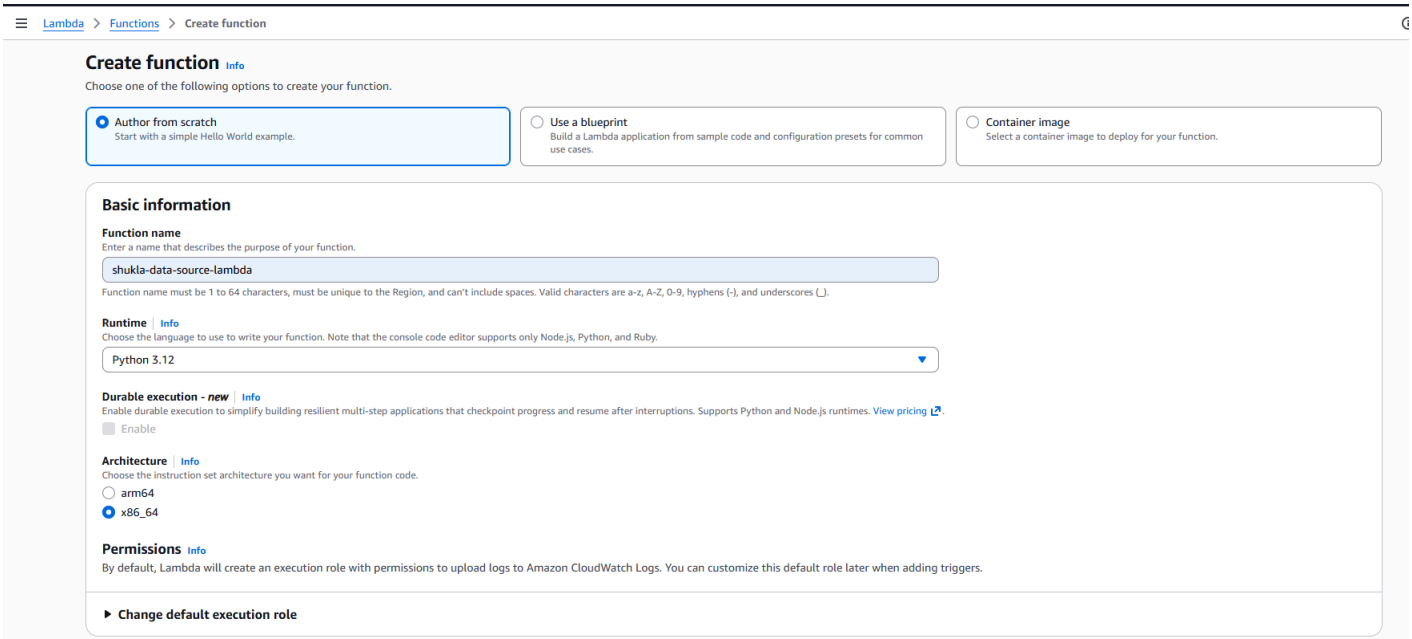


Lets create a Lambda and Assign the role and

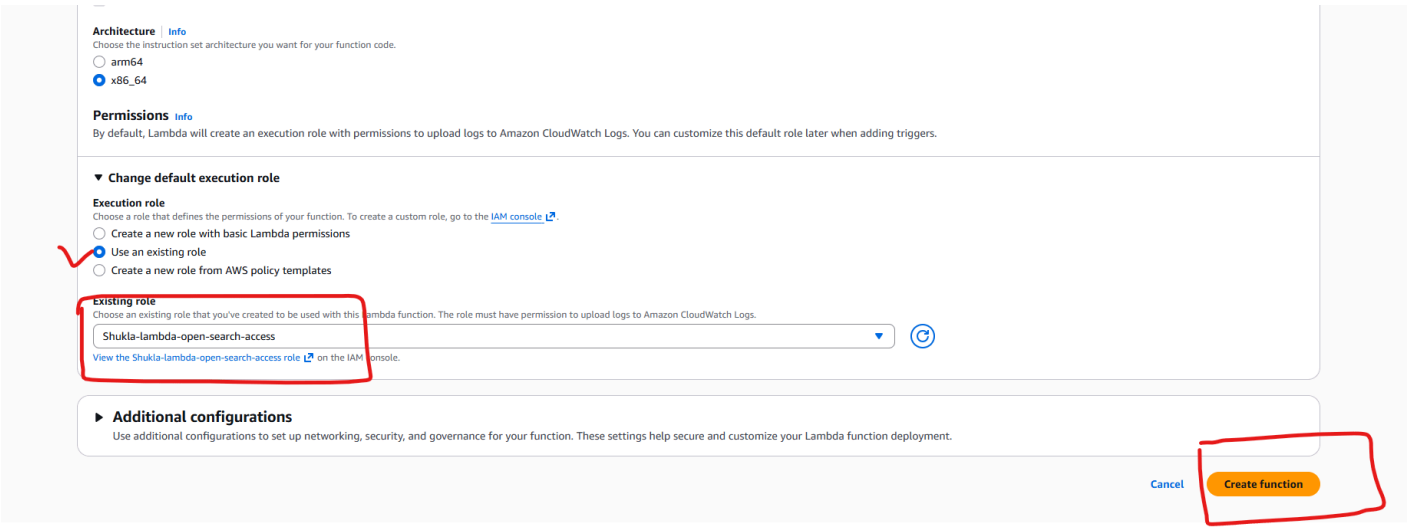
permission to it.

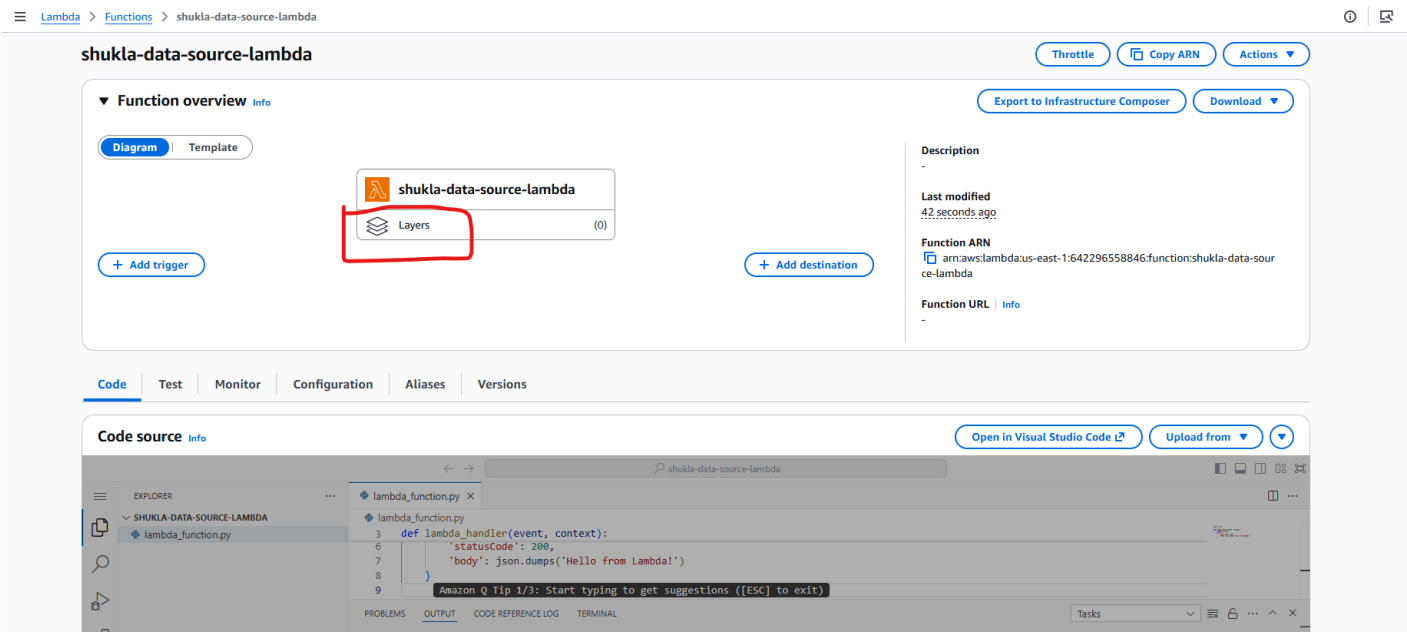
Go to AWS console and search form Lambda





use previously created Role to assign to this lambda. and click createe

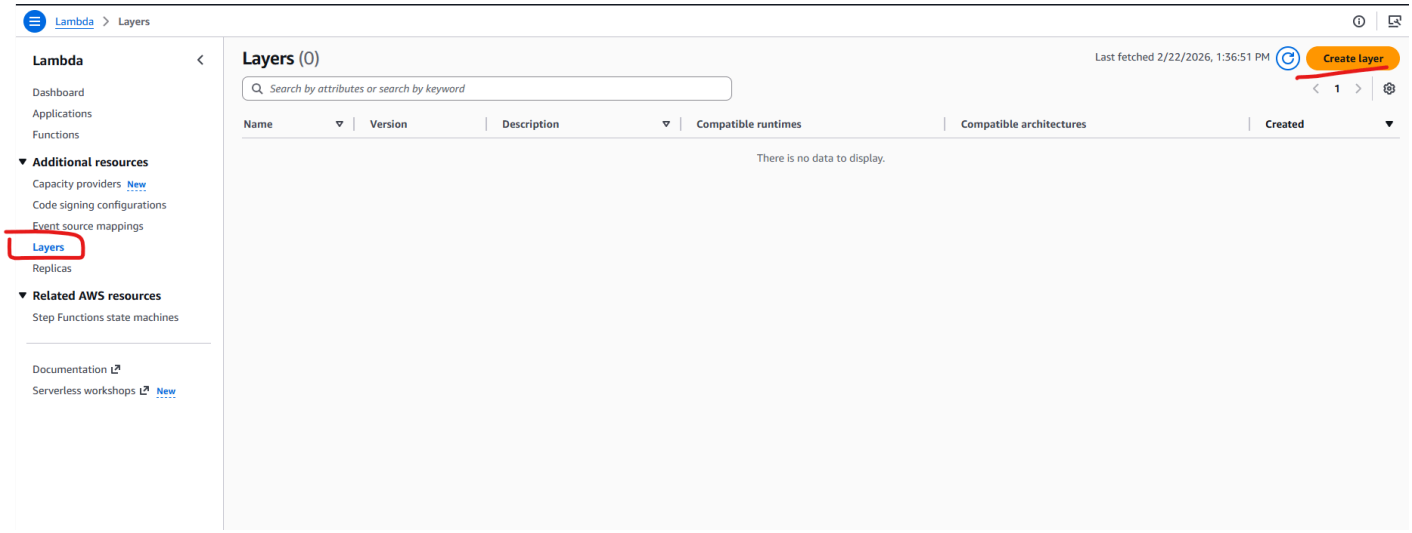


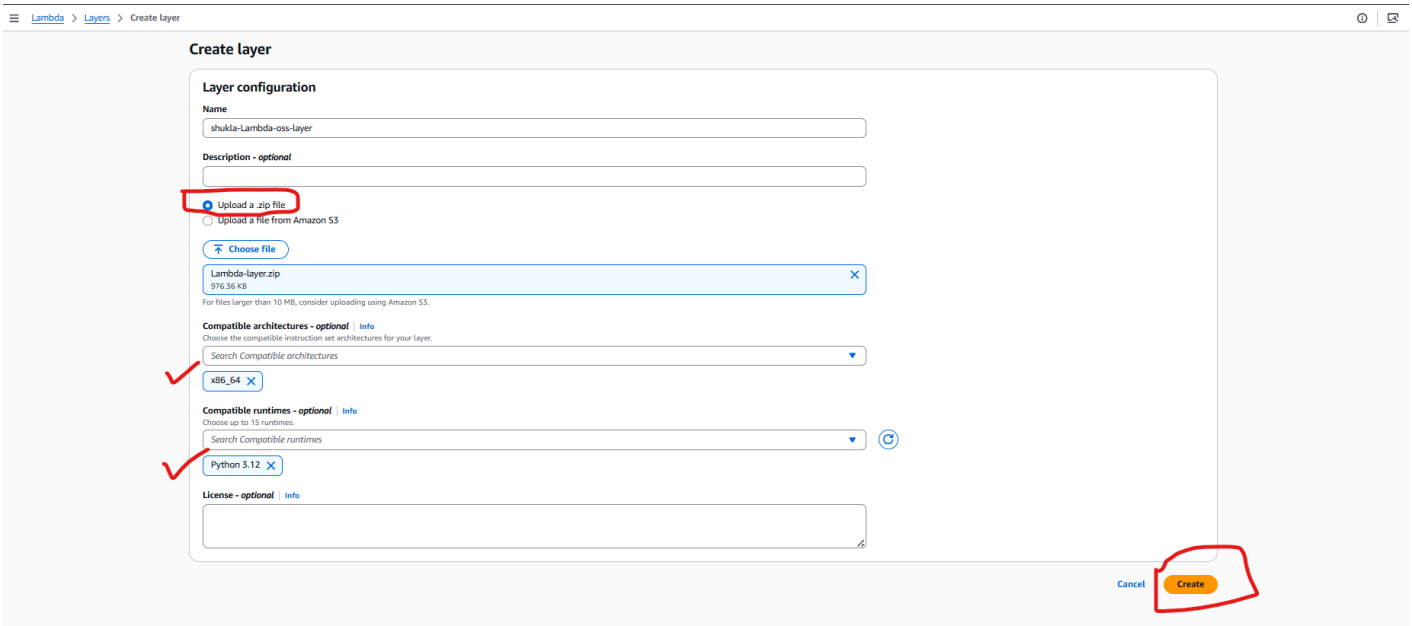


Our lambda function is created in python , this will generate some random data and insert that data in previously created open search collection.

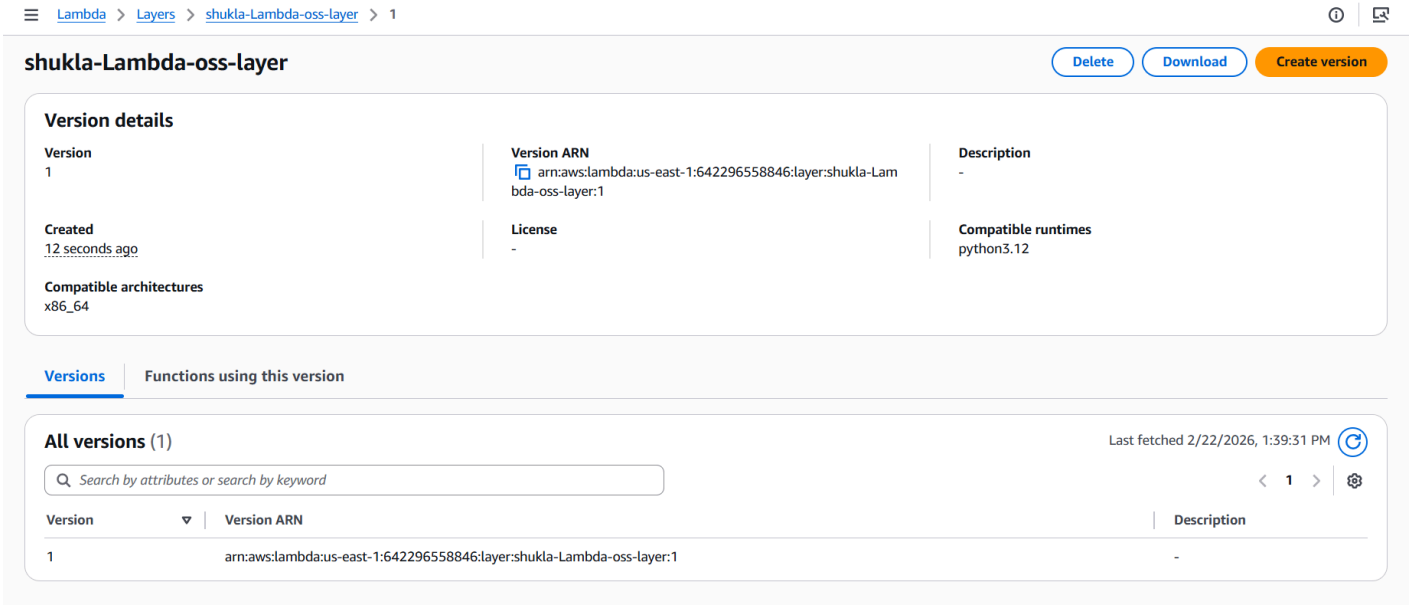
This code has some dependency. these dependencies can not be directly input in the code. AWS lambda use provides a concept of layers to manage it.

Lets create a layers that can be used in this lambda



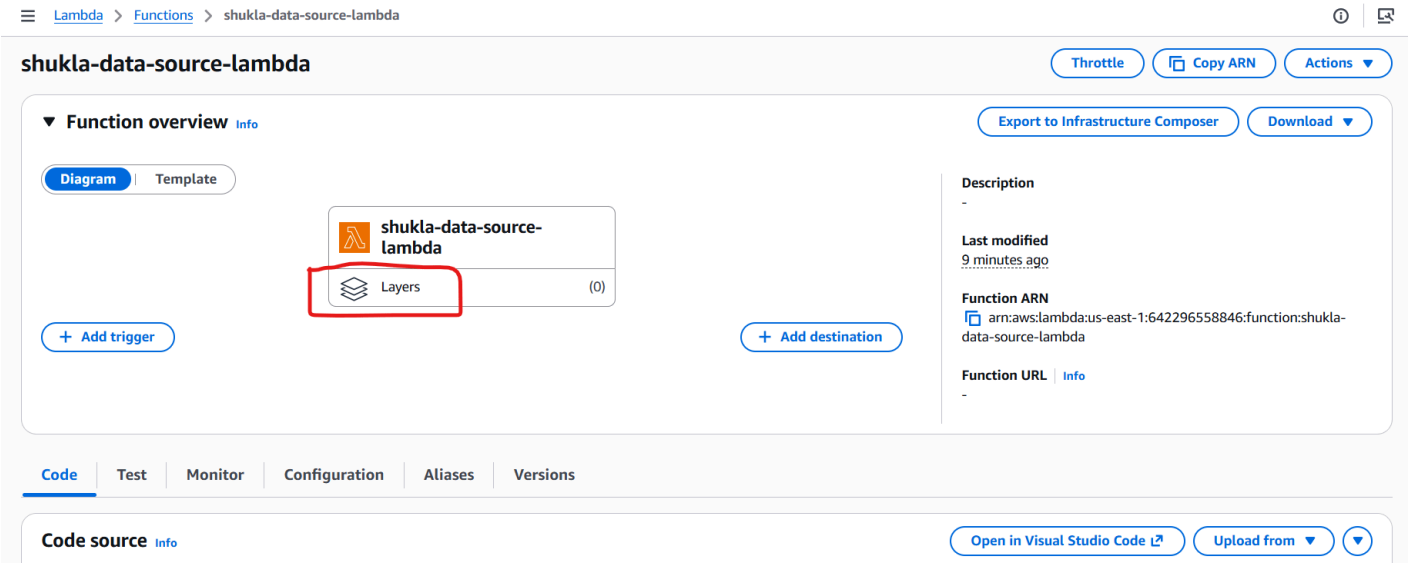


Layer is now ready to reuse.

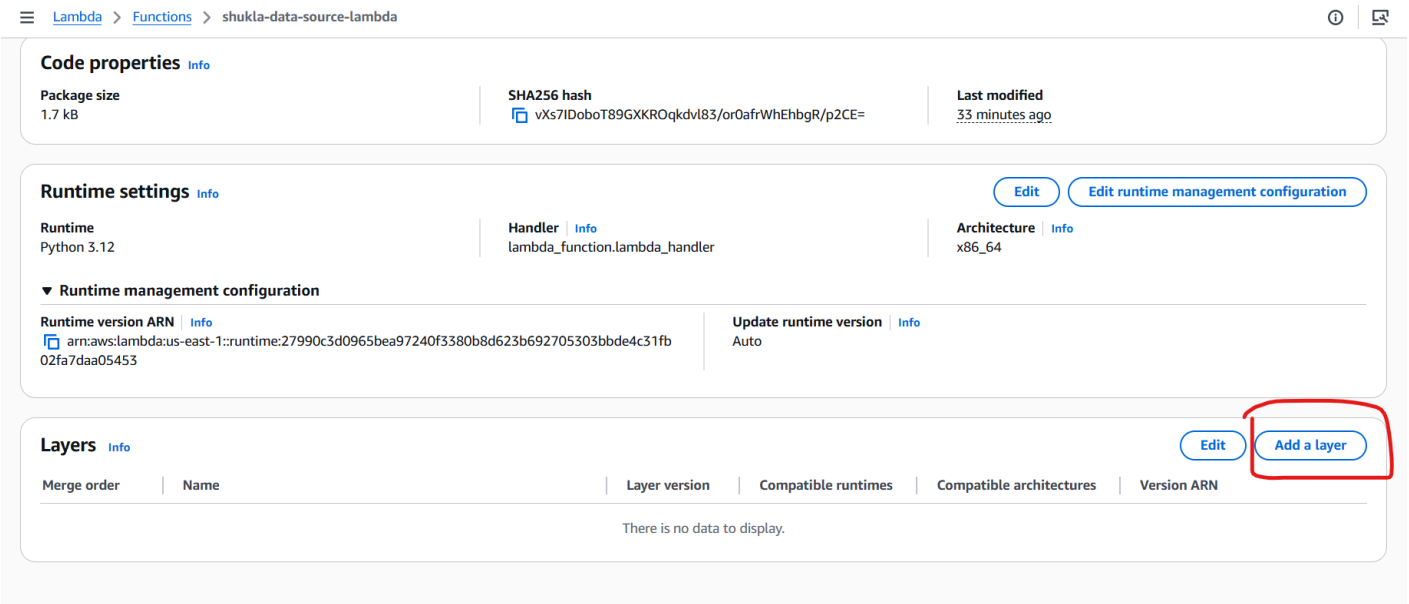


Lets add this layer in our lambda

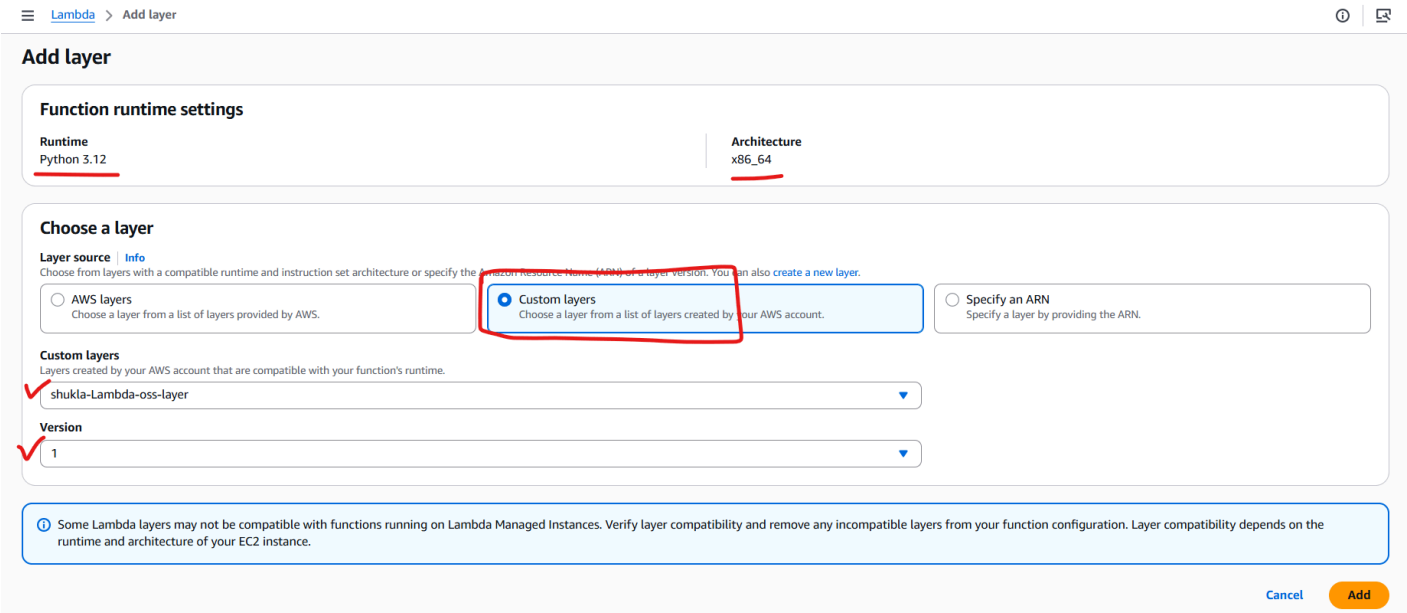
Go to the Lambda and click on layers.



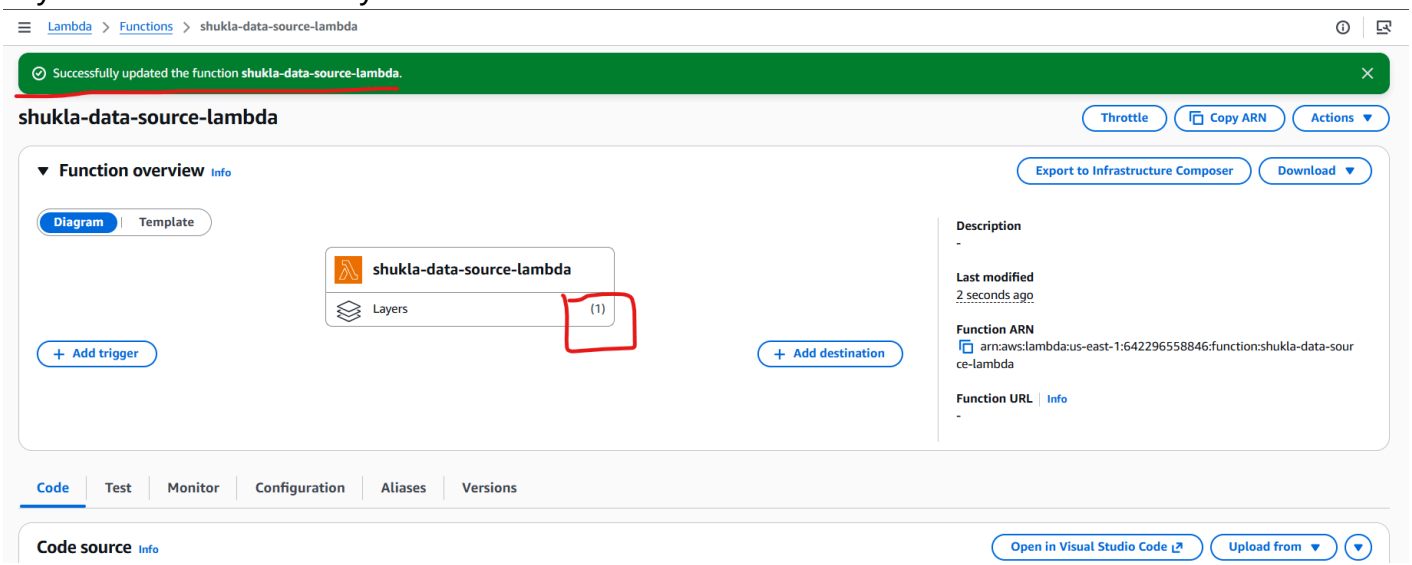
Click on Add a layer.



Select the previously added layer from the drop down. Click Add



Layer is now successfully Added to the Lambda.



Lets deploy code to this lambda.

Here is the lambda code. make sure you change the Open search endpoint with your.

plaintext

```
import json
import boto3
import requests
import uuid
import random
from datetime import datetime, timedelta
from requests_aws4auth import AWS4Auth
from botocore.exceptions import BotoCoreError, NoCredentialsError
from requests.exceptions import RequestException, Timeout, ConnectionError

# ---------- CONFIG ----------
region = "us-east-1"
service = "aoss"
host = "Open Search Host url here"
index_name = "demo-index"

headers = {"Content-Type": "application/json"}

# ---------- MESSAGE GENERATOR ----------
def generate_message(service):
    messages = {
        "PaymentService": [
            "Payment processed successfully",
            "Card authorization failed",
            "Refund initiated"
        ],
        "OrderService": [
            "Order created",
            "Order validation failed",
            "Inventory reserved"
        ],
        "AuthService": [
            "User login success",
            "Invalid token detected",
            "Session expired"
        ]
    }
    return random.choice(messages.get(service, ["Unknown event"]))

# ---------- DATA GENERATION ----------
def generate_records(count=500):

    services = ["PaymentService", "OrderService", "AuthService"]
    logs = []
```

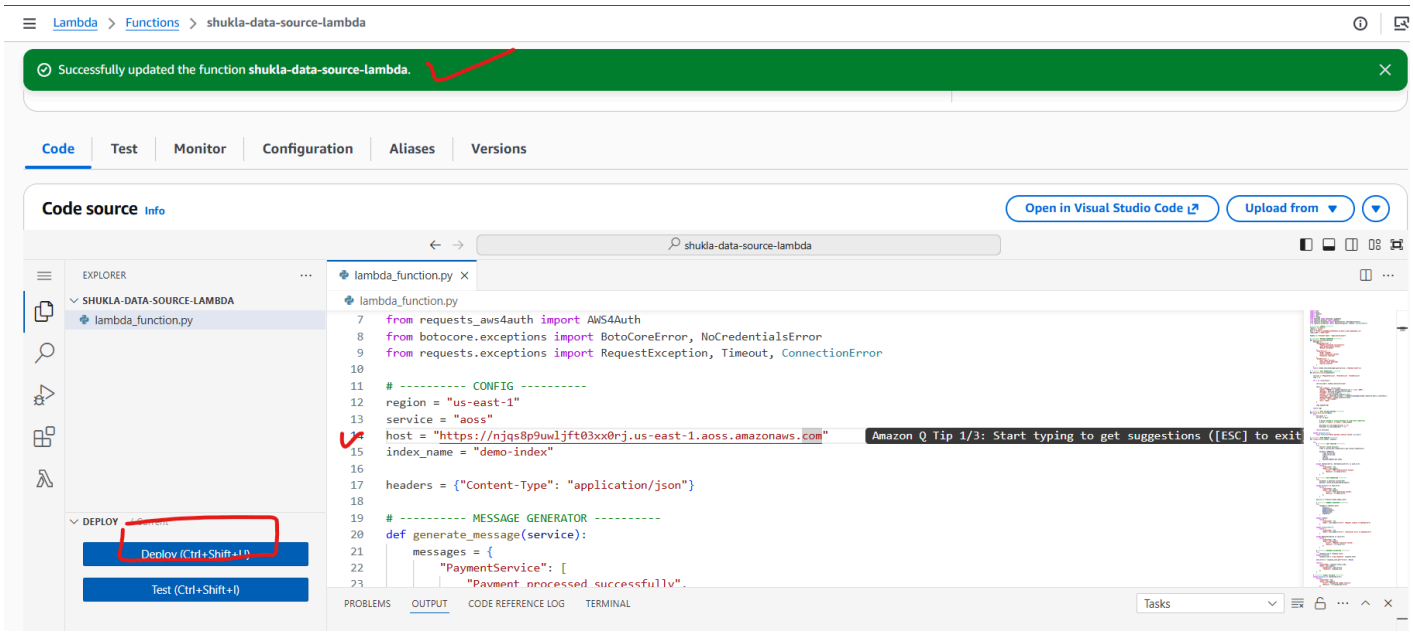
```
    for i in range(count):

        service_name = random.choice(services)

        log = {
            "ServiceName": service_name,
            "Level": "ERROR" if random.randint(0,10) > 7 else "INFO",
            "Message": generate_message(service_name),
            "TraceId": str(uuid.uuid4()),
            "UserId": f"user-{random.randint(1,10)}",
            "Timestamp": (datetime.utcnow() -
timedelta(seconds=random.randint(0,600))).isoformat(),
            "ResponseTimeMs": random.randint(50,800),
            "source": "aws-lambda",
            "env": "demo"
        }

        logs.append(log)

    return logs

# ---------- BULK PAYLOAD BUILDER ----------
def build_bulk_payload(docs):
    try:
        bulk_data = ""
        for doc in docs:

            # Using TraceId as unique document id (log-style ingestion)
            action = {"index": {"_index": index_name}}

            bulk_data += json.dumps(action) + "\n"
            bulk_data += json.dumps(doc) + "\n"

        return bulk_data

    except Exception as e:
        raise Exception(f"Bulk payload creation failed: {str(e)}")

# ---------- MAIN HANDLER ----------
def lambda_handler(event, context):

    try:
        # ---------- AUTH CREATION ----------
        try:
```

```
        session = boto3.Session()
        creds = session.get_credentials().get_frozen_credentials()

        awsauth = AWS4Auth(
            creds.access_key,
            creds.secret_key,
            region,
            service,
            session_token=creds.token
        )

    except (BotoCoreError, NoCredentialsError) as auth_error:
        return {
            "statusCode": 500,
            "body": json.dumps({
                "error": "AWS authentication failed",
                "details": str(auth_error)
            })
        }

    # ---------- DATA GENERATION ----------
    try:
        documents = generate_records(500)
        payload = build_bulk_payload(documents)

    except Exception as data_error:
        return {
            "statusCode": 500,
            "body": json.dumps({
                "error": "Data generation failed",
                "details": str(data_error)
            })
        }

    bulk_url = f"{host}/{index_name}/_bulk"

    # ---------- REQUEST EXECUTION ----------
    try:
        response = requests.post(
            bulk_url,
            auth=awsauth,
            headers=headers,
            data=payload,
            timeout=30
        )
```

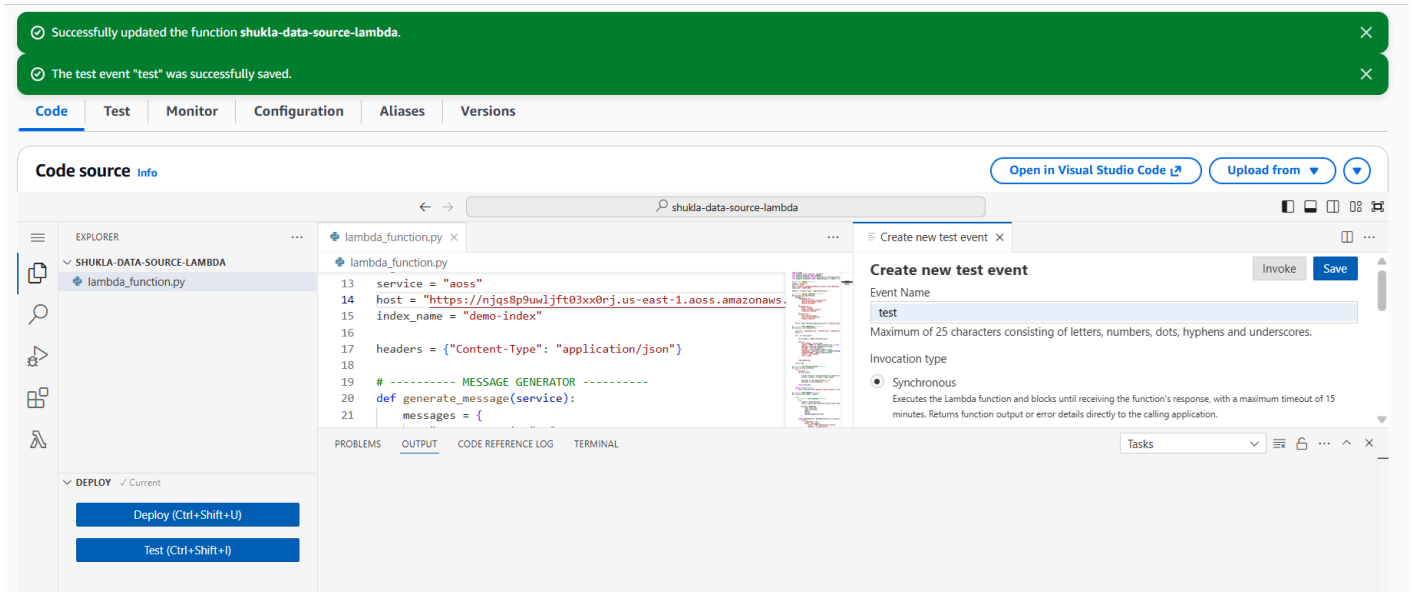
```
    except Timeout:
        return {
            "statusCode": 504,
            "body": json.dumps({"error": "Request timeout to OpenSearch"})
        }

    except ConnectionError:
        return {
            "statusCode": 503,
            "body": json.dumps({"error": "Connection error to OpenSearch"})
        }

    except RequestException as req_error:
        return {
            "statusCode": 500,
            "body": json.dumps({
                "error": "Request execution failed",
                "details": str(req_error)
            })
        }

    # ---------- RESPONSE VALIDATION ----------
    try:
        response_json = response.json()
    except Exception:
        response_json = {"raw_response": response.text}

    bulk_errors = response_json.get("errors", False)

    return {
        "statusCode": response.status_code,
        "body": json.dumps({
            "bulkErrors": bulk_errors,
            "response": response_json
        })
    }

# ---------- GLOBAL FALLBACK ----------
except Exception as unexpected_error:
    return {
        "statusCode": 500,
        "body": json.dumps({
            "error": "Unexpected lambda failure",
            "details": str(unexpected_error)
```

```
            })
        }
```

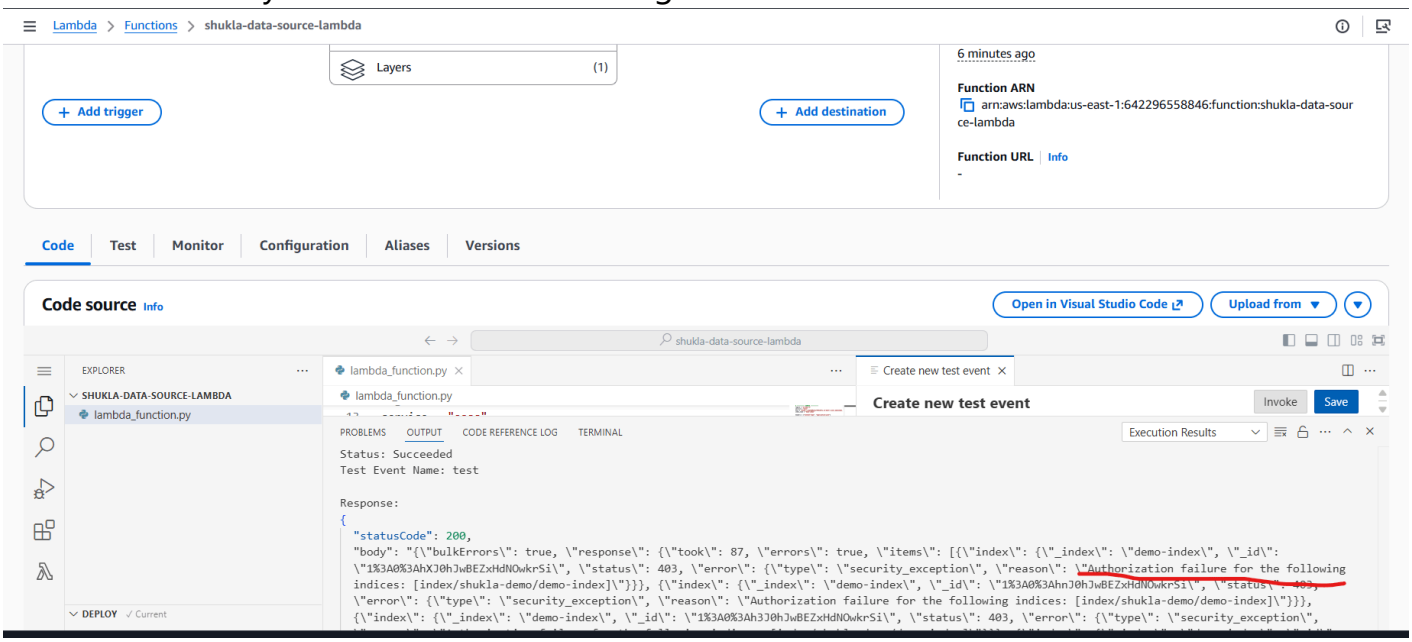
Lets deploy the lambda.



Lets create a test event .



Run the test and you will see that it showing autorization failure.



Reason for this failure it. that lambda is not able to connect with AWS open search console.

Lets change the data access policy of AWS open search console and add the "ARN of the role created earlier"

Go to the open search console and access the data access policy.

Amazon OpenSearch Service > Serverless: Collections > shukla-demo

Network — Manage network access

Associated Policy	Access type	Resource type
auto-shukla-demo	Public	Access to Opensearch endpoints Access to Opensearch Dashboards

Encryption

KMS ARN
AWS owned key

Data access — Manage data access

Data access policies apply to collections and indexes, and control a user's access to the data in these resource.

Associated policy
shukla-data-access-policy

Data lifecycle policy (1) — Create data lifecycle policy

The data lifecycle refers to the period of time that data exists. You can create a data lifecycle policy to specify how long data will be stored before being deleted.

Search

Data lifecycle policy	Affected index	Data retention
Default (no policy)	All Indexes	Unlimited (Never delete)

Amazon OpenSearch Service > Serverless: Security > Data access control > shukla-data-access-policy > editAccessPolicy

Edit access policy Info

Name and description

Access policy name
shukla-data-access-policy

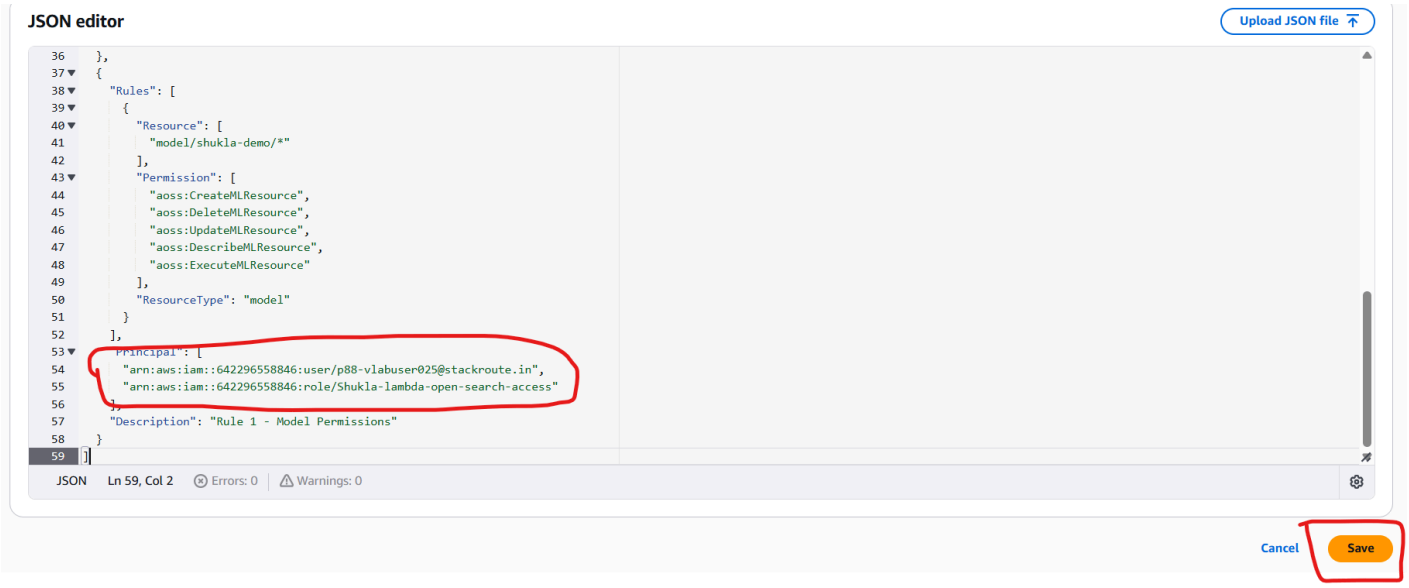
Description - *optional*
Enter description

Select policy definition method
- Visual editor
- JSON

JSON editor — Upload JSON file

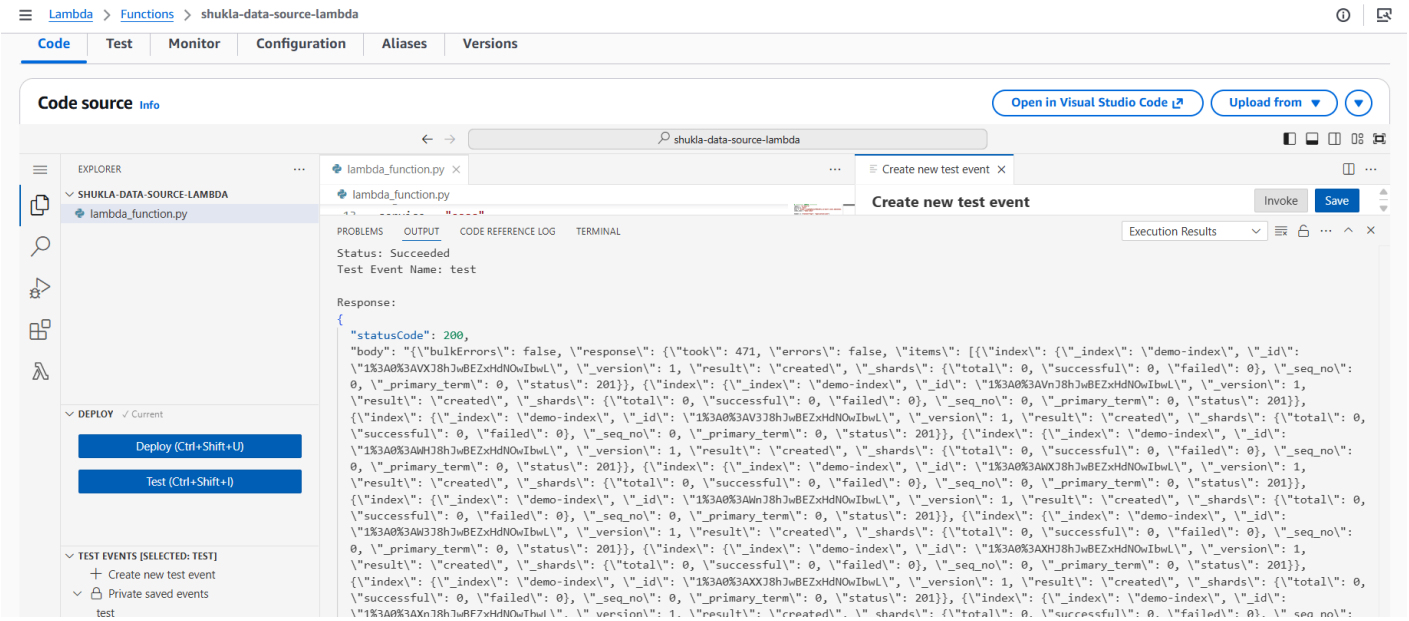
```
[
  {
    "Rules": [
      {
        "Resource": [
          "collection/shukla-demo"
        ],
        "Permission": [
          "aoss:CreateCollectionItems",
          "aoss:DeleteCollectionItems",
```

Add the Role arn and click save.



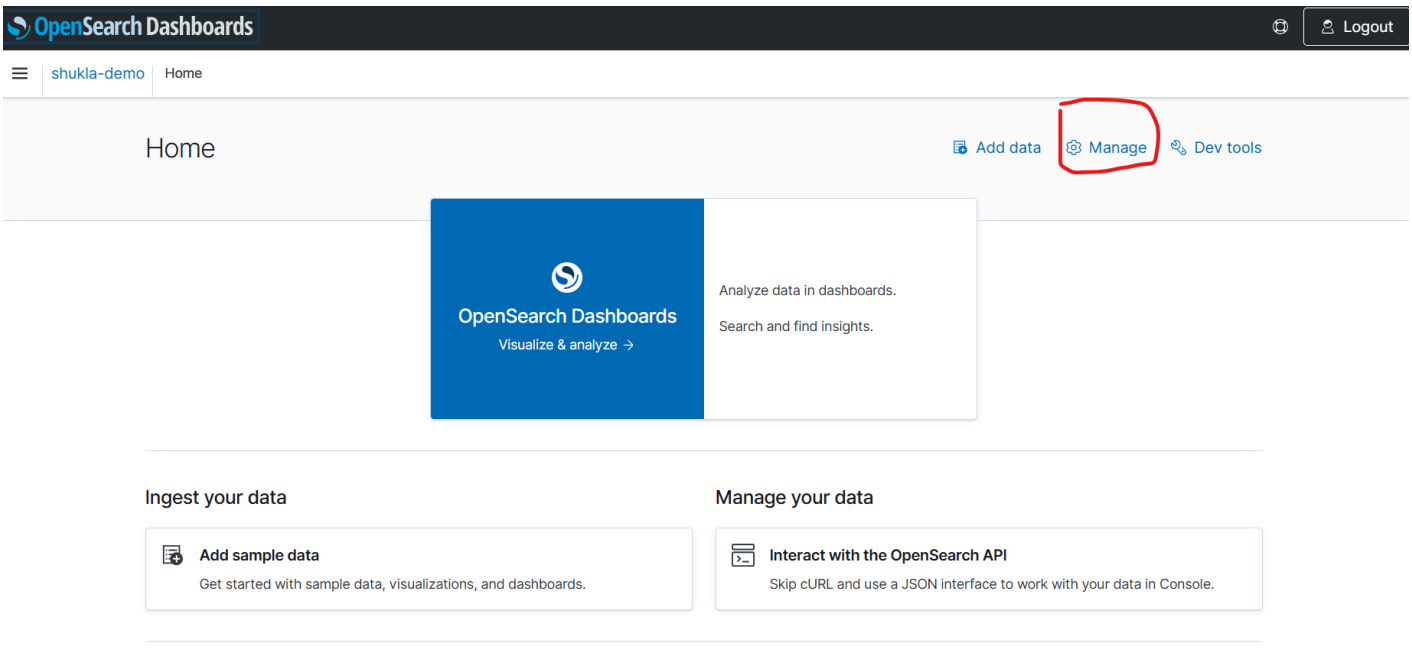
Wait for 1-2 minute to policy get in effect.

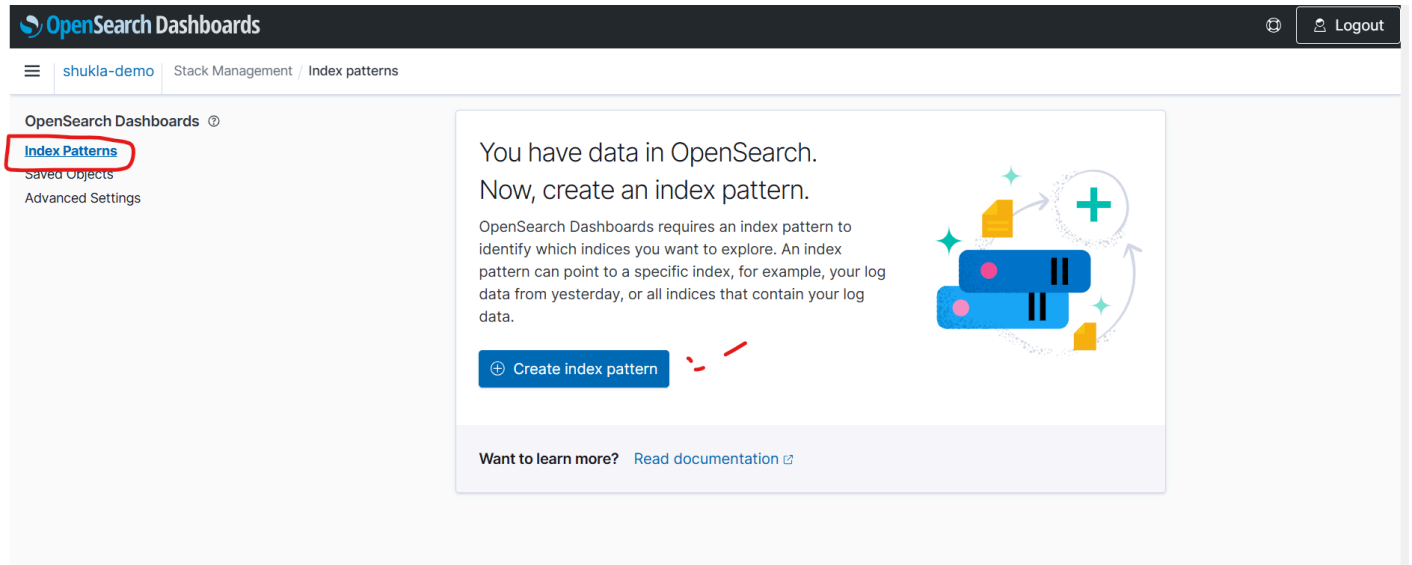
Lets execute the Lambda function again. we can see the success.

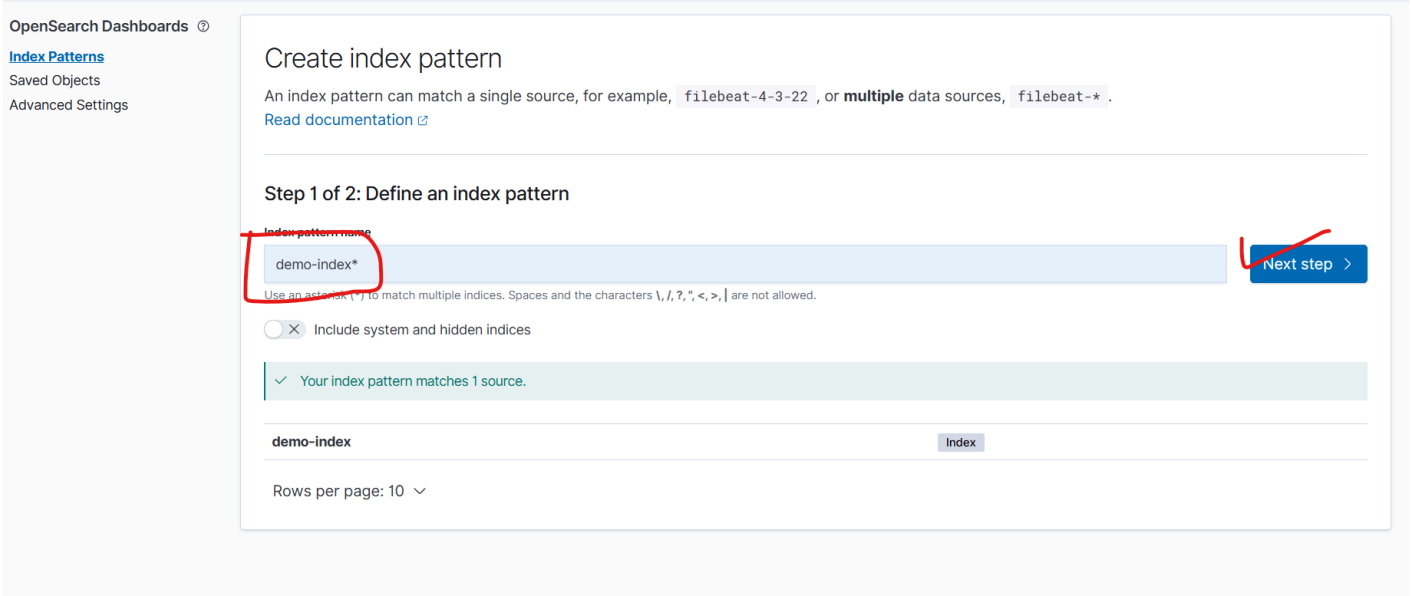


Lets create a dashboard of the inserted data for

visualization.







shukla-demo | Stack Management / Index patterns / Create index pattern

OpenSearch Dashboards

Index Patterns
Saved Objects
Advanced Settings

Create index pattern

An index pattern can match a single source, for example, `filebeat-4-3-22`, or **multiple** data sources, `filebeat-*`.
Read documentation

Step 2 of 2: Configure settings

Specify settings for your **demo-index*** index pattern.

Select a primary time field for use with the global time filter.

Time field Refresh

Timestamp

Show advanced settings

Back | Create index pattern

shukla-demo | Stack Management / Index patterns / demo-index*

OpenSearch Dashboards

Index Patterns
Saved Objects
Advanced Settings

demo-index*

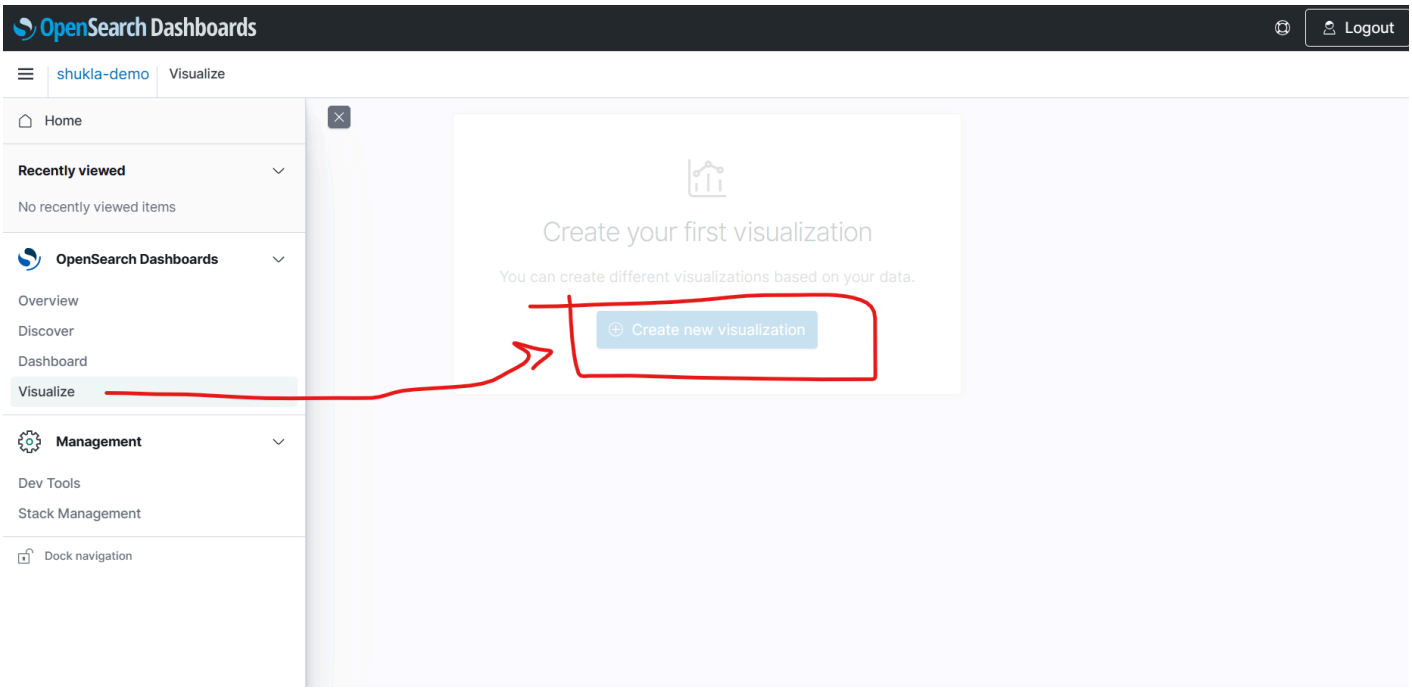
Time field: 'Timestamp'

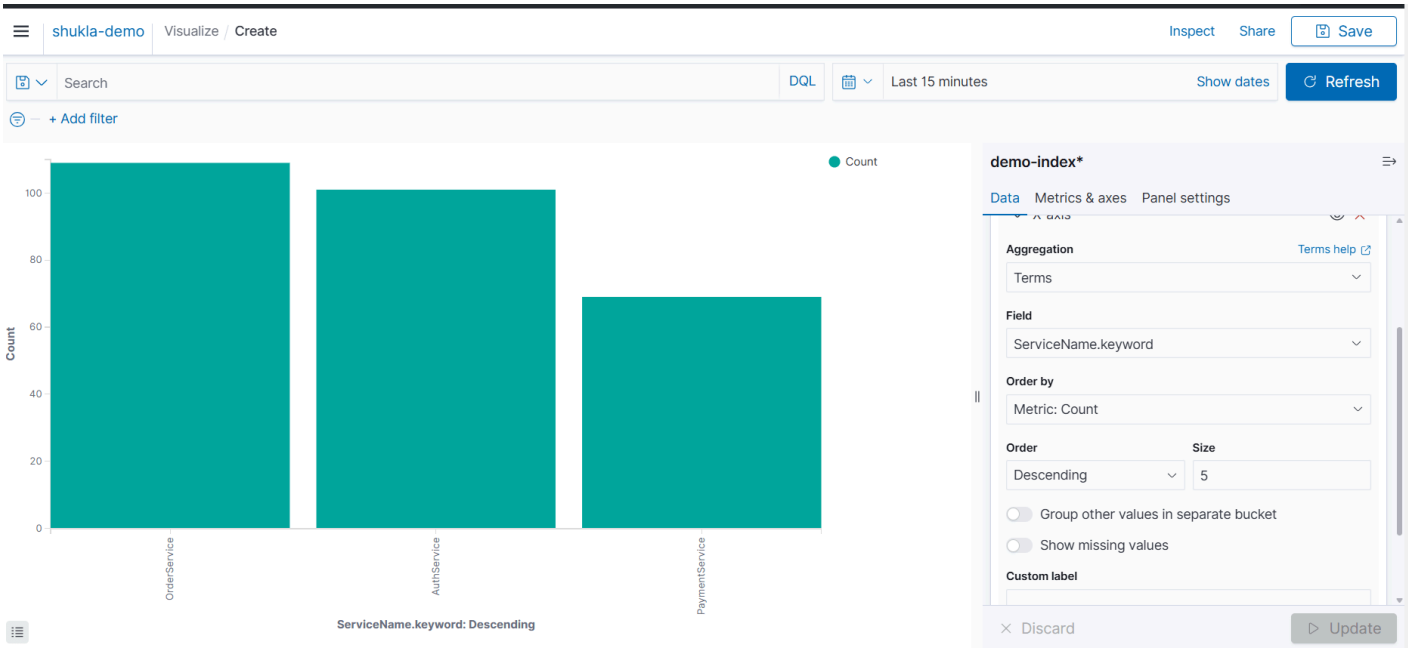
This page lists every field in the **demo-index*** index and the field's associated core type as recorded by OpenSearch. To change a field type, use the OpenSearch Mapping API

Fields (21) | Scripted fields (0) | Source filters (0)

Search | All field types

Name	Type	Format	Searchable	Aggregatable	Excluded
Level	string		•		
Level.keyword	string		•	•	
Message	string		•		
Message.keyword	string		•	•	
ResponseTimeMs	number		•	•	
ServiceName	string		•		
ServiceName.keyword	string		•	•	


OpenSearch Dashboards
Logout
shukla-demo
Visualize
Home
Recently viewed
No recently viewed items
OpenSearch Dashboards
Overview
Discover
Dashboard
Visualize
Management
Dev Tools
Stack Management
Dock navigation
Create your first visualization
You can create different visualizations based on your data.
Create new visualization


shukla-demo
Visualize / Create
Inspect
Share
Save
Search
DQL
Last 15 minutes
Show dates
Refresh
+ Add filter
Count
100
80
60
40
20
0
OrderService
AuthService
PaymentService
ServiceName.keyword: Descending
demo-index*
Data
Metrics & axes
Panel settings
Aggregation
Terms help
Terms
Field
ServiceName.keyword
Order by
Metric: Count
Order
Descending
Size
5
Group other values in separate bucket
Show missing values
Custom label
Discard
Update